

Alzikrayat Photo Sharing Application

Architecture & Implementation Report

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Project Overview

Alzikrayat is a photo-sharing web application built from scratch in PHP, with no backend framework, following a custom MVC (Model-View-Controller) architecture. Registered users can upload photos, browse a shared gallery, comment on photos, and manage their own uploads, including a custom server-side image filter (grayscale/sepia) built with the PHP GD library.

Architecture & Project Structure

The project follows a hand-written MVC structure with a manual, regex-based routing engine (no framework router):

- **config/** — shared PDO database connection (Singleton pattern)
- **core/** — base Model, Controller, and Router classes
- **controllers/** — AuthController, PhotoController, CommentController, HomeController
- **models/** — User, Photo, Comment (database access layer)
- **views/** — PHP page templates, styled with Bootstrap 5
- **public/** — entry point (index.php), CSS/JS assets, and uploaded images

The Router class converts each **{param}** placeholder in a route pattern into a regex capture group, matches it against the incoming URL, and dispatches the request directly to the matching Controller action.

Database Design

The MySQL schema has three normalized tables: **users**, **photos**, and **comments**. Both photos and comments carry foreign keys back to users (and photos, in the case of comments) with **ON DELETE CASCADE**, so removing a user automatically removes their photos and comments. Every query in the Model layer uses PDO prepared statements with named or positional placeholders.

Main Features

- **Authentication:** registration and login with bcrypt password hashing, PHP sessions, and a 7-day "last login" cookie.
- **Photo Gallery:** upload, view, and delete photos, with ownership checked at both the Controller and SQL levels.
- **Comments:** users can comment on photos; comments post instantly via AJAX, with a fallback to a normal form submit.

- **Triple-layer Validation:** input is checked with HTML5 attributes, client-side JavaScript, and server-side PHP.
- **Security:** PDO prepared statements protect against SQL injection, and htmlspecialchars() on all dynamic output protects against XSS.

Novelty / Advanced Features

As the project's novelty component, a custom server-side image filter engine was built directly on PHP's GD library, offering **Grayscale** and **Sepia** filters with no external image-processing API. Before a filter is applied to a photo for the first time, an untouched backup of the original file is saved automatically, and a one-click **Restore Original** action lets the owner revert to it at any time.

Conclusion

Alzikrayat meets the project's architectural requirements: a hand-written MVC structure, a regex-based manual router, raw parameterized SQL with no ORM, secure session-based authentication, ownership-validated CRUD operations on photos, and a novelty image-filter feature layered on top of the core requirements.